

Electric Circuits



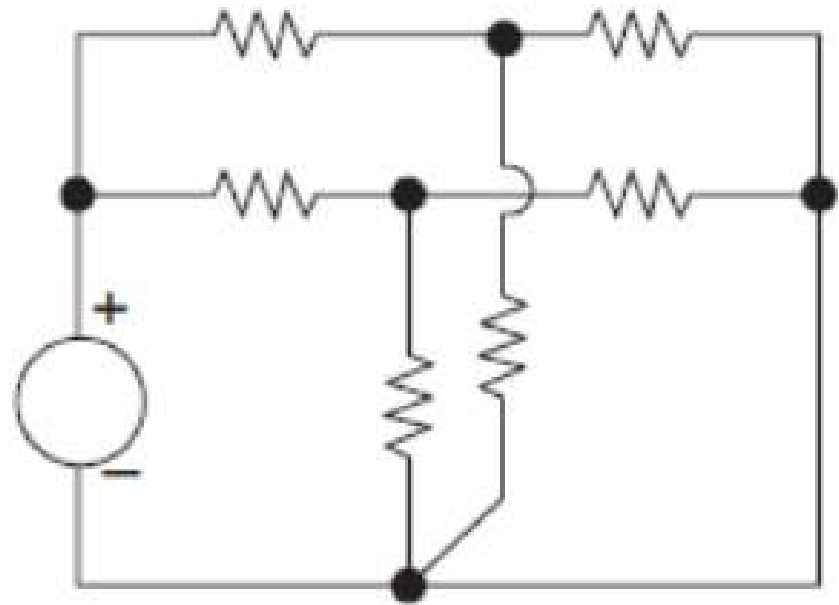
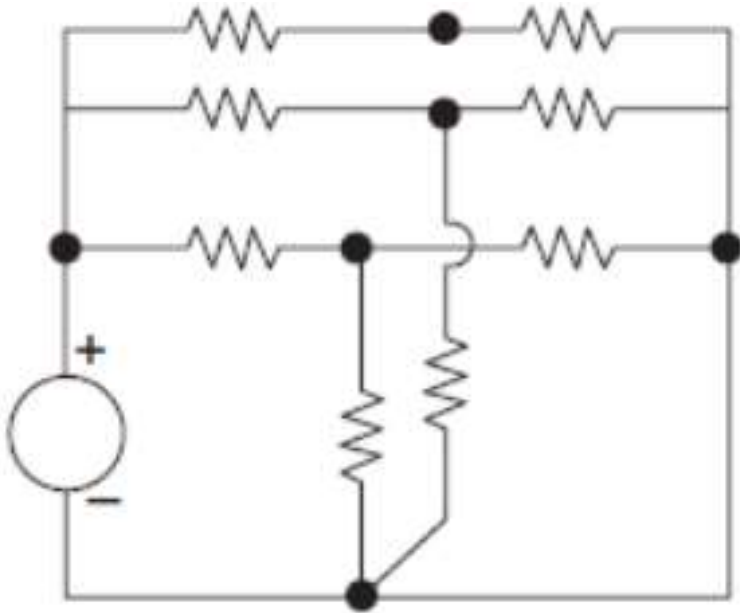
*Mesh Analysis: Circuits with
Voltage & Current Sources*

Tentative Lecture Plan

L	LDay-1	LDay-2	Topics	T	TDay
1	FRI, 04 AUG	MON, 07 AUG	Basic Definitions, Voltage & Current Sources, KCL & KVL	1	THU, 10 AUG
2	TUE, 08 AUG	FRI, 11 AUG	Nodal Analysis: Independent and Dependent Current Sources	2	SAT, 19 AUG
3	MON, 14 AUG	THU, 17 AUG	Nodal Analysis: Current and Voltage Sources		
4	FRI, 18 AUG	MON, 21 AUG	Mesh Analysis: Voltage and Current Sources	3	THU, 24 AUG
5	TUE, 22 AUG	FRI, 25 AUG	Network Theorems: Superposition Theorem		
SUN, 27 AUG : Quiz-1 : Lectures 1-5					
6	FRI, 26 AUG	MON, 28 AUG	Network Theorems: Thevenin's and Norton's Theorem	4	THU, 31 AUG
7	TUE, 29 AUG	FRI, 04 SEP	Network Theorems: Maximum Power Transfer Theorem	5	WED, 06 SEP
8	TUE, 05 SEP	FRI, 08 SEP	Series RL and RC Circuits		
9	SAT, 09 SEP	MON, 11 SEP	Series RL, RC, RLC Circuits Response to Constant, Sinusoid and Exponential Input	6	THU, 14 SEP
10	TUE, 12 SEP	FRI, 15 SEP	Concept of Phasor, AC Circuits, AC Power	R	SAT, 16 SEP
WED, 20 SEP: MidSem : Lectures 1 – 10					

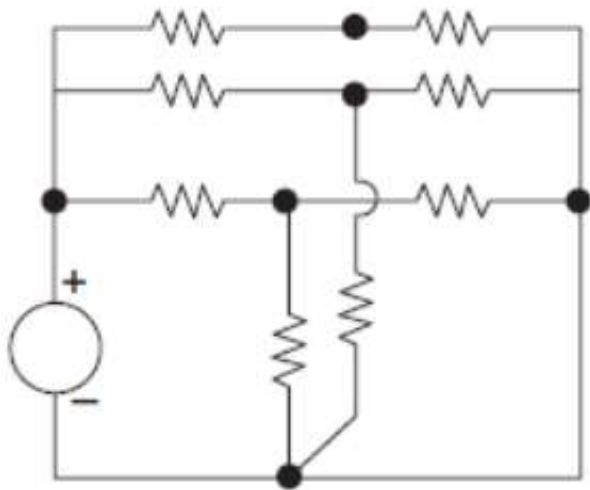
Planar and Non-Planar Networks

A planar network is one that can be drawn on a plane without any branch crossing over another

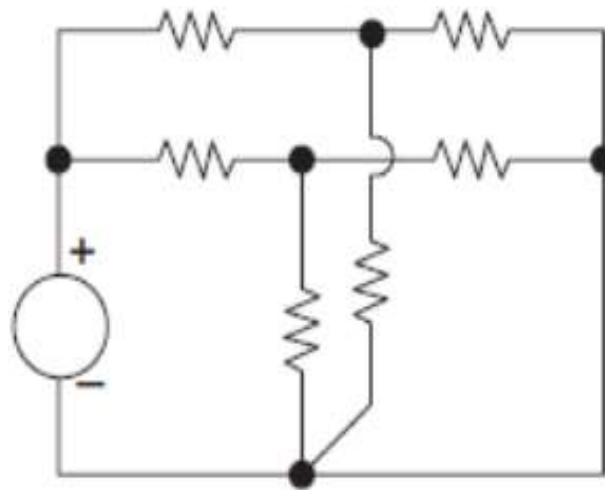


Planar and Non-Planar Networks

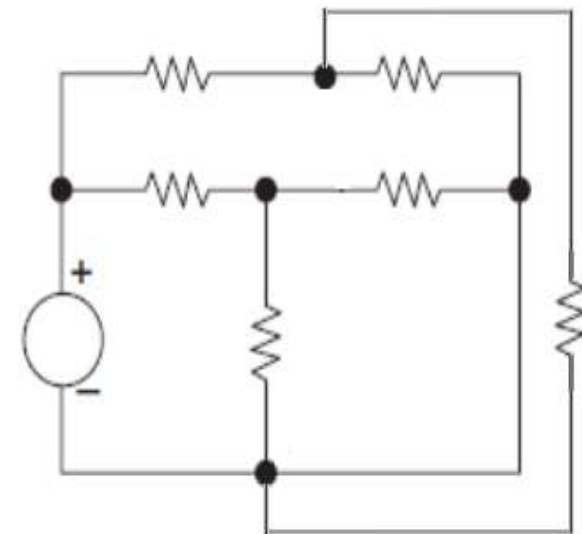
A planar network is one that can be drawn on a plane without any branch crossing over another



Non-Planar?

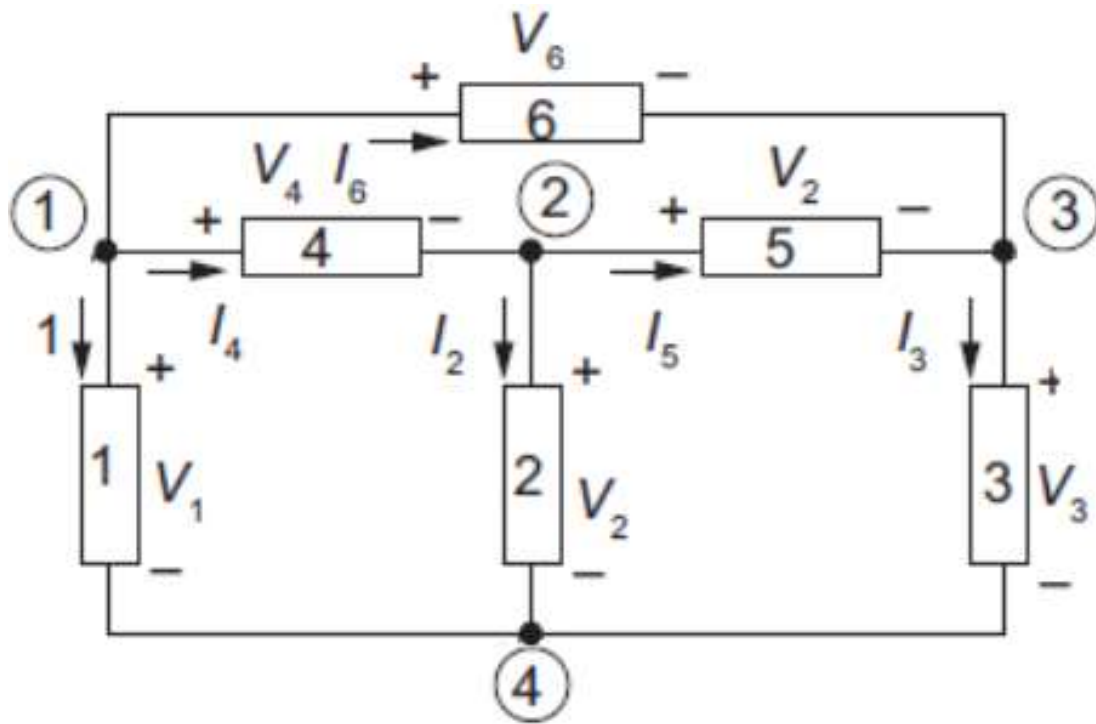


Planar



Loops and Meshes

- **Loop:** A closed path comprising of branch elements and nodes
- **Mesh:** A loop in a planar circuit that does not contain any other loop in it

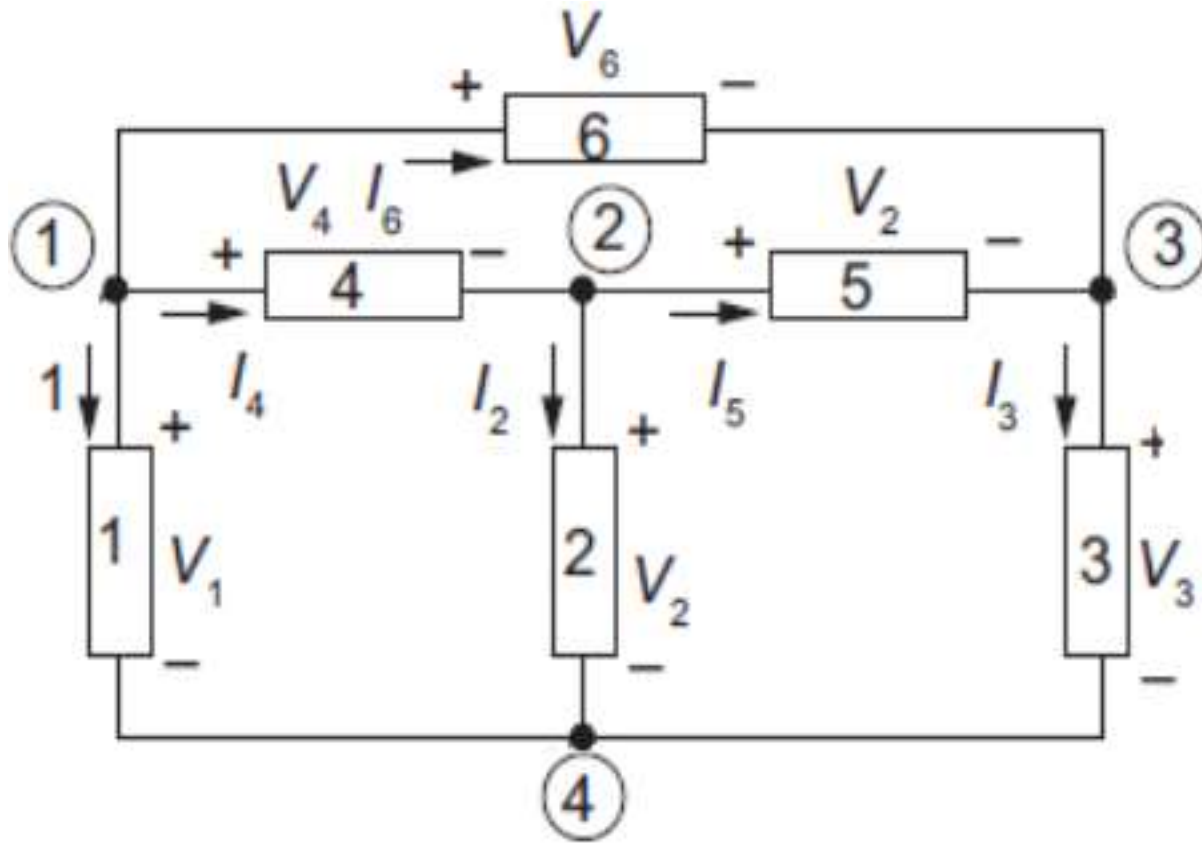


The branches are numbered

Identify the Loops and Meshes

Loops and Meshes

- **Loop:** A closed path comprising of branch elements and nodes
- **Mesh:** A loop in a planar circuit that does not contain any other loop in it



$L_1 (M_1) : 4-6-5$

$L_2 (M_2) : 4-2-1$

$L_3 (M_3) : 5-3-2$

$L_4 : 6-3-1$

$L_5 : 6-3-2-4$

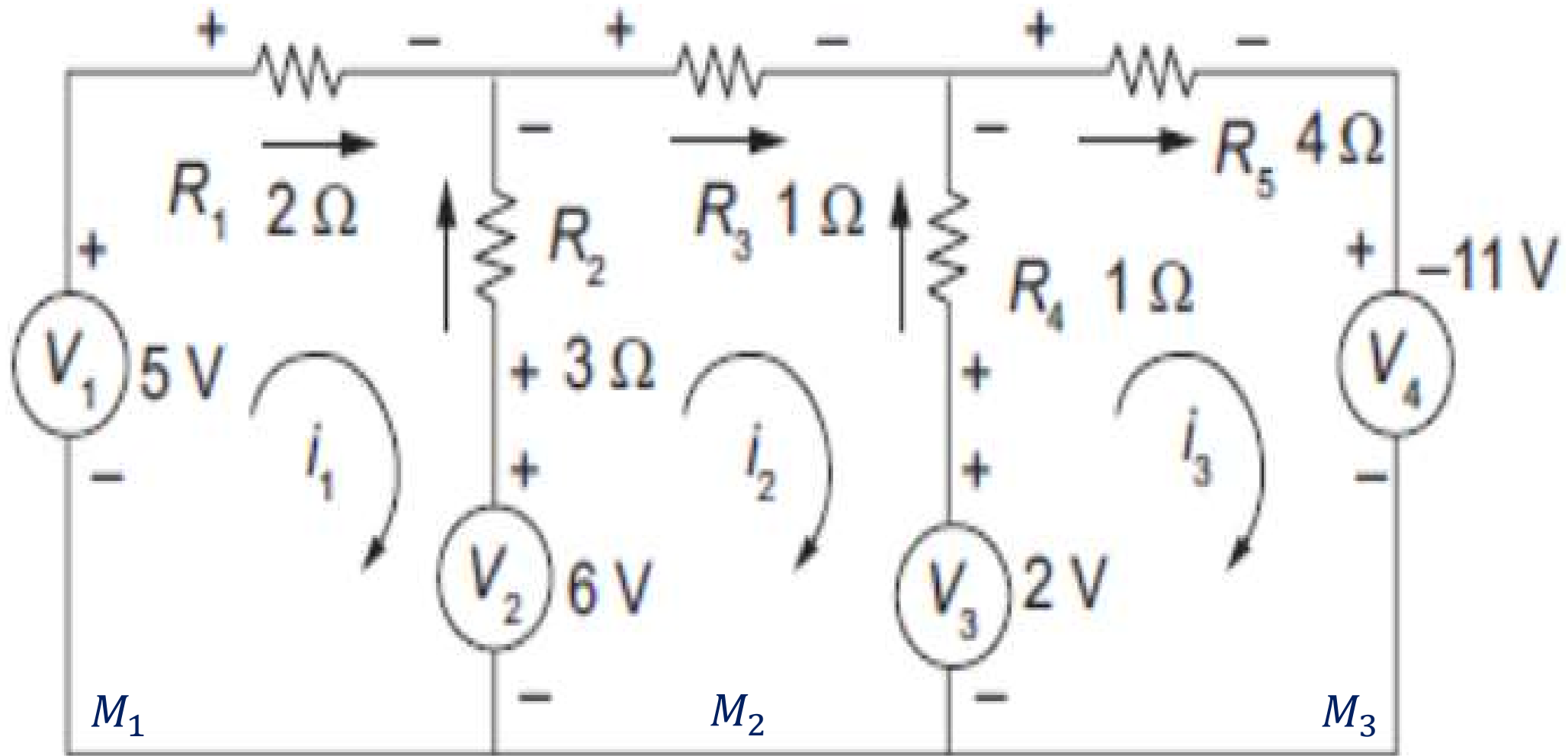
$L_6 : 6-5-2-1$

$L_7 : 4-5-3-1$

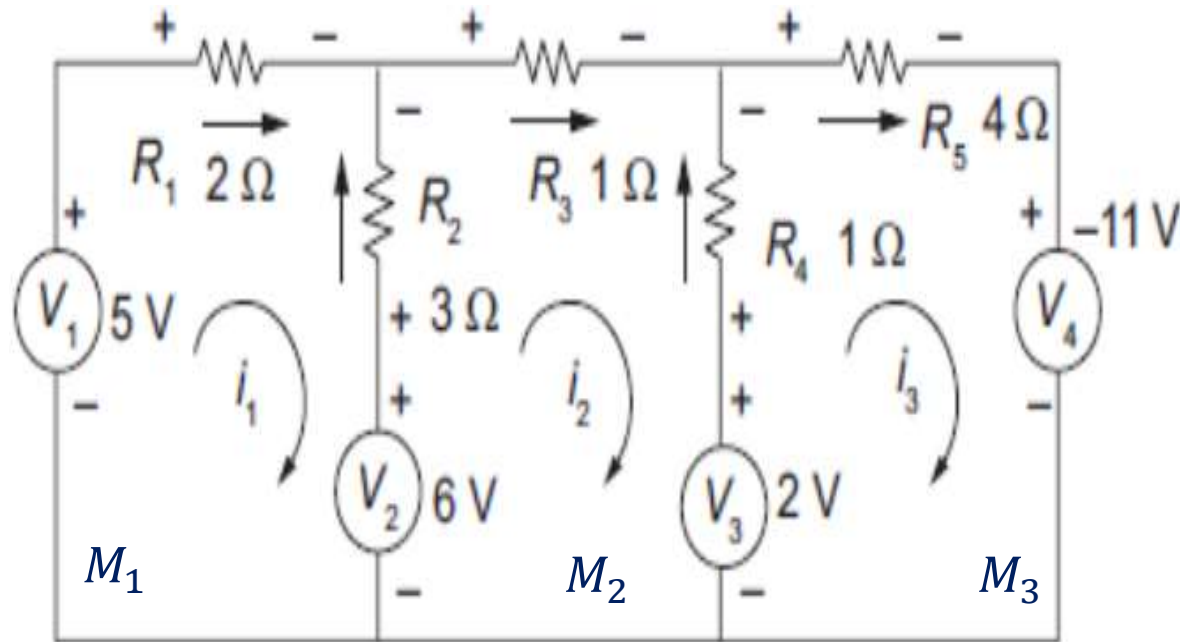
Circuit Analysis Equations

- A circuit with n nodes, b branches and l loops will have n KCL equations, l KVL equations and b element equations involving $2b$ element variables
- Any $(n - 1)$ KCL equations of n will be linearly independent
- Only $(b - n + 1)$ KVL equations will be linearly independent
- An arbitrary set of $(b - n + 1)$ KVL equations drawn from a set of l such equations need not form a linearly independent set
- There will be exactly $(b - n + 1)$ meshes
- The KVL equations for $(b - n + 1)$ meshes in a planar circuit will form a linearly independent set of equations

Mesh Analysis with Independent Voltage Source



Mesh Analysis with Independent Voltage Source

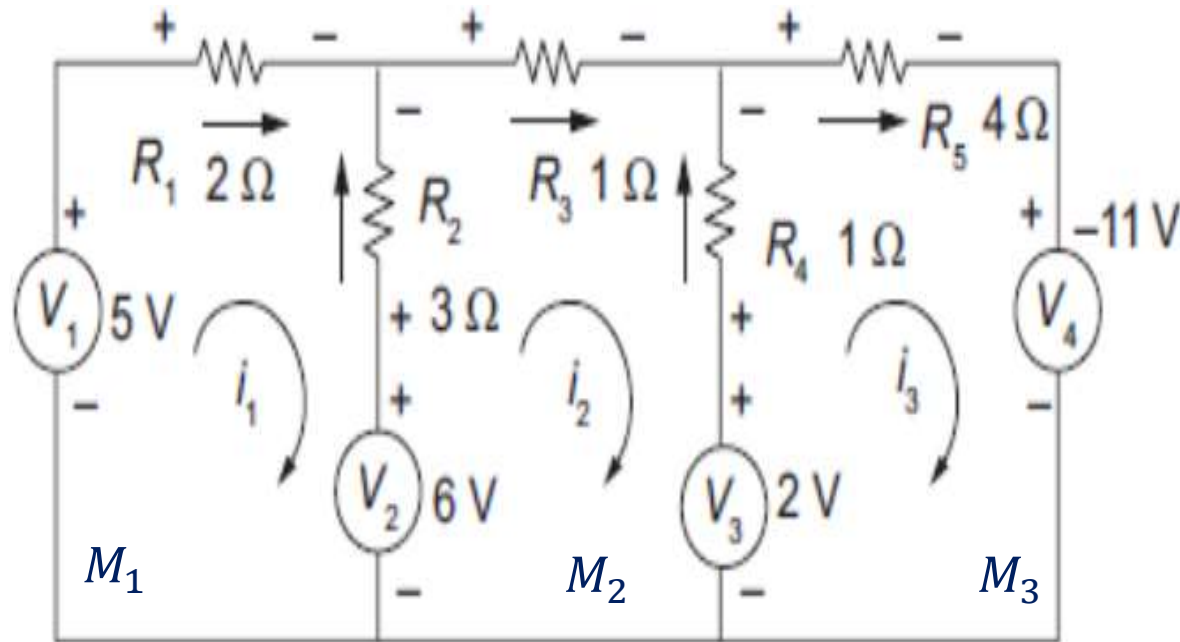


KVL for Mesh **1**

$$-V_1 + R_1 i_1 - R_2 (i_2 - i_1) + V_2 = 0$$

$$(R_1 + R_2) i_1 - R_2 i_2 = V_1 - V_2$$

Mesh Analysis with Independent Voltage Source

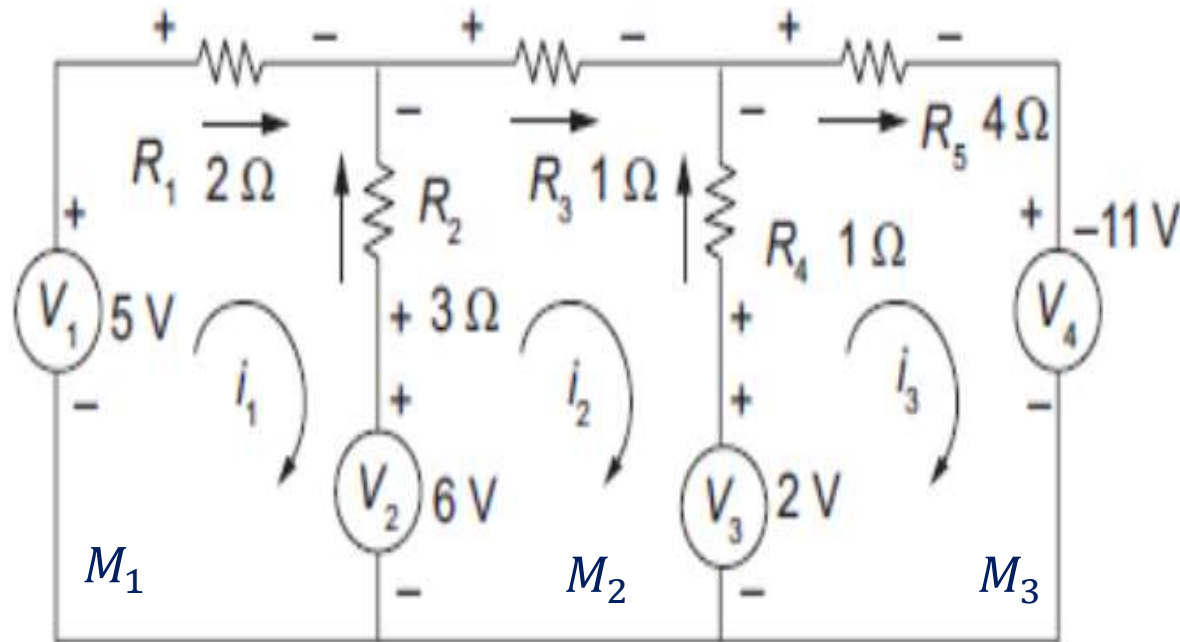


KVL for Mesh 2

$$-V_2 + R_2(i_2 - i_1) + R_3 i_2 - R_4(i_3 - i_2) + V_3 = 0$$

$$-R_2 i_1 + (R_2 + R_3 + R_4) i_2 - R_4 i_3 = V_2 - V_3$$

Mesh Analysis with Independent Voltage Source



KVL for Mesh **3**

$$-V_3 + R_4(i_3 - i_2) + R_5 i_3 + V_4 = 0$$

$$-R_4 i_2 + (R_4 + R_5) i_3 = V_3 - V_4$$

Mesh Analysis with Independent Voltage Source

$$(R_1 + R_2)i_1 - R_2i_2 = V_1 - V_2$$

$$-R_2i_1 + (R_2 + R_3 + R_4)i_2 - R_4i_3 = V_2 - V_3$$

$$-R_4i_2 + (R_4 + R_5)i_3 = V_3 - V_4$$

$$\begin{bmatrix} R_1 + R_2 & -R_2 & 0 \\ -R_2 & R_2 + R_3 + R_4 & -R_4 \\ 0 & -R_4 & R_4 + R_5 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix}$$

Mesh Analysis with Independent Voltage Source

$$\begin{bmatrix} R_1 + R_2 & -R_2 & 0 \\ -R_2 & R_2 + R_3 + R_4 & -R_4 \\ 0 & -R_4 & R_4 + R_5 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix}$$

Mesh Resistance Matrix

Input Matrix

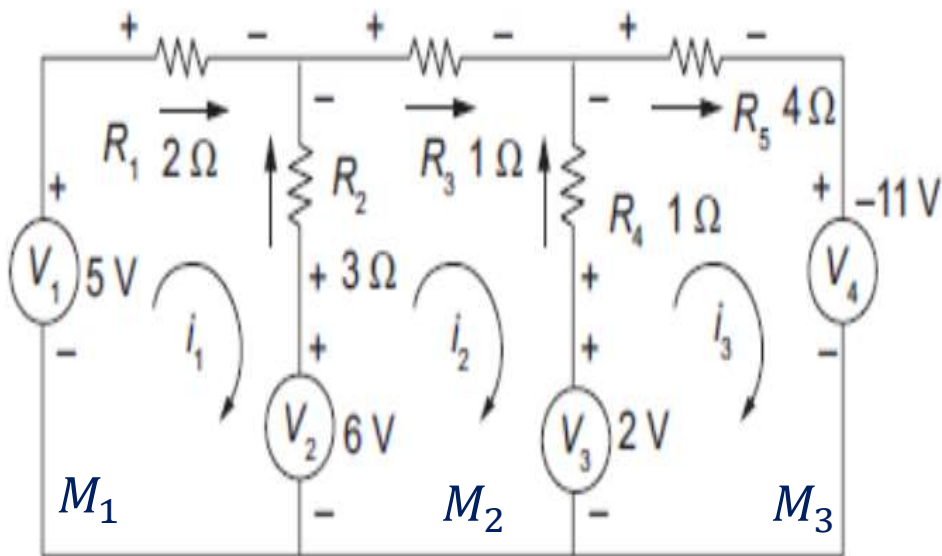
$$\begin{matrix} [Z]_{m \times m} & [I]_{m \times 1} & = & [D]_{m \times n_{ivs}} & [U]_{n_{ivs} \times 1} \\ \text{Mesh Current Vector} & & & & \text{Independent Source Vector} \end{matrix}$$

$m = b - n + 1$: Number of Meshes

n_{ivs} : Number of Independent Voltage Sources

Mesh Analysis with Independent Voltage Source

$$\begin{bmatrix} R_1 + R_2 & -R_2 & 0 \\ -R_2 & R_2 + R_3 + R_4 & -R_4 \\ 0 & -R_4 & R_4 + R_5 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix}$$

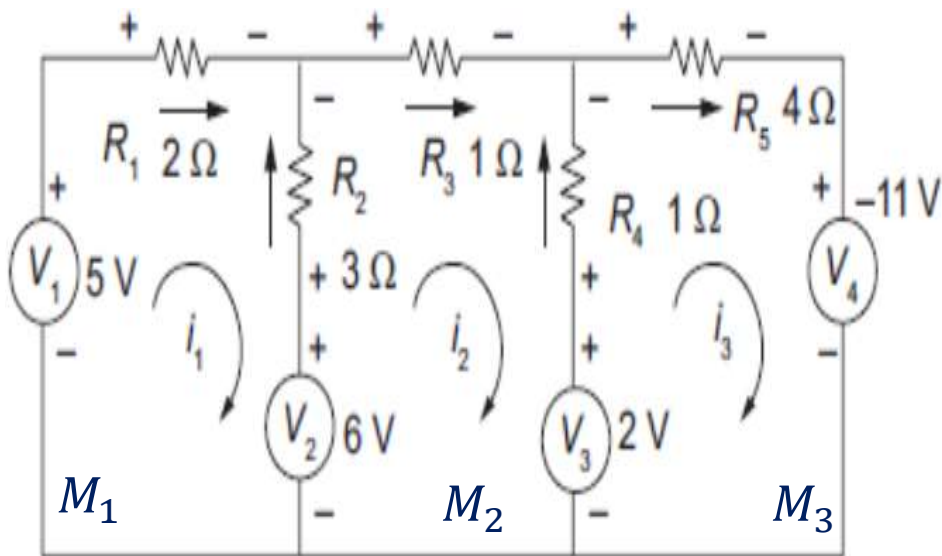


$Z[i][i]$: Sum of All Resistances
Connected to Mesh i

$Z[i][j] = Z[j][i]$: Negative of Sum
of All Resistances Connected
(Shared) between Mesh i and
Mesh j

Mesh Analysis with Independent Voltage Source

$$\begin{bmatrix} R_1 + R_2 & -R_2 & 0 \\ -R_2 & R_2 + R_3 + R_4 & -R_4 \\ 0 & -R_4 & R_4 + R_5 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix}$$



$D[i][j] = 0$: Source $U[j]$ Not Connected to Mesh i

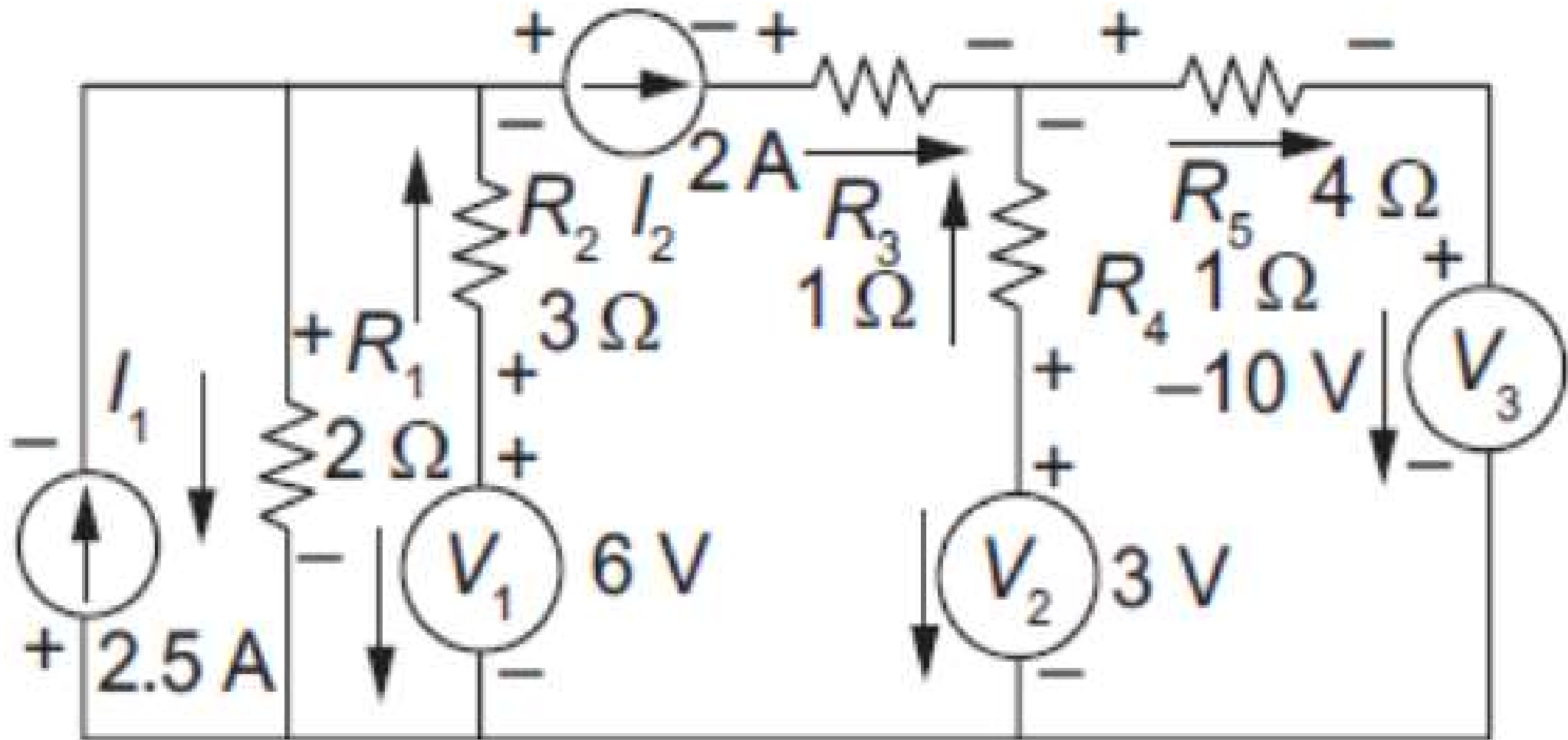
$D[i][j] = 1$: Source $U[j]$ provides a Voltage Rise in Mesh i

$D[i][j] = -1$: Source $U[j]$ provides a Voltage Drop in Mesh i

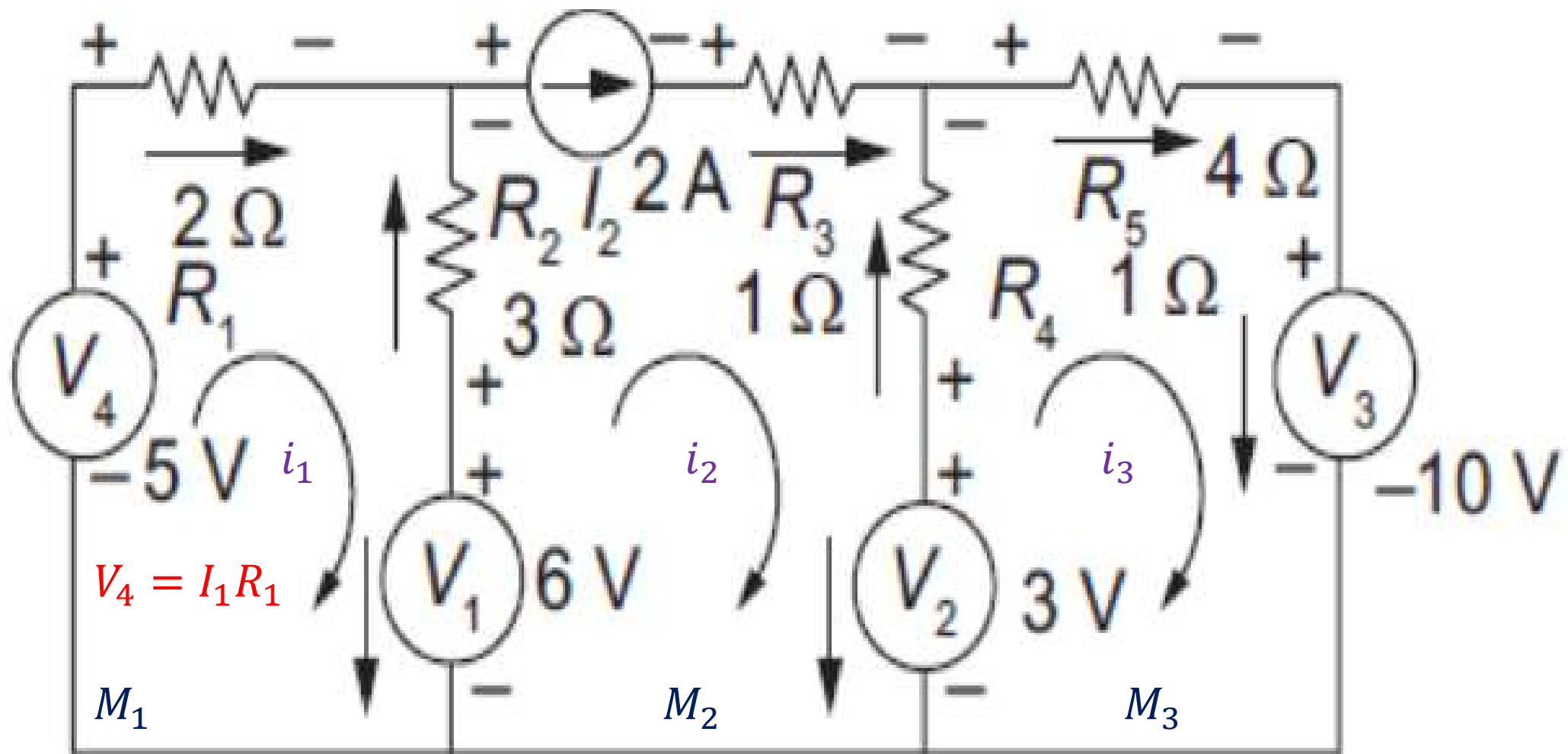
Presence of Independent Current Source

- Independent Current Source in parallel with a resistor
- Independent Current Source in series with another element
- Independent Current Source shared between two meshes

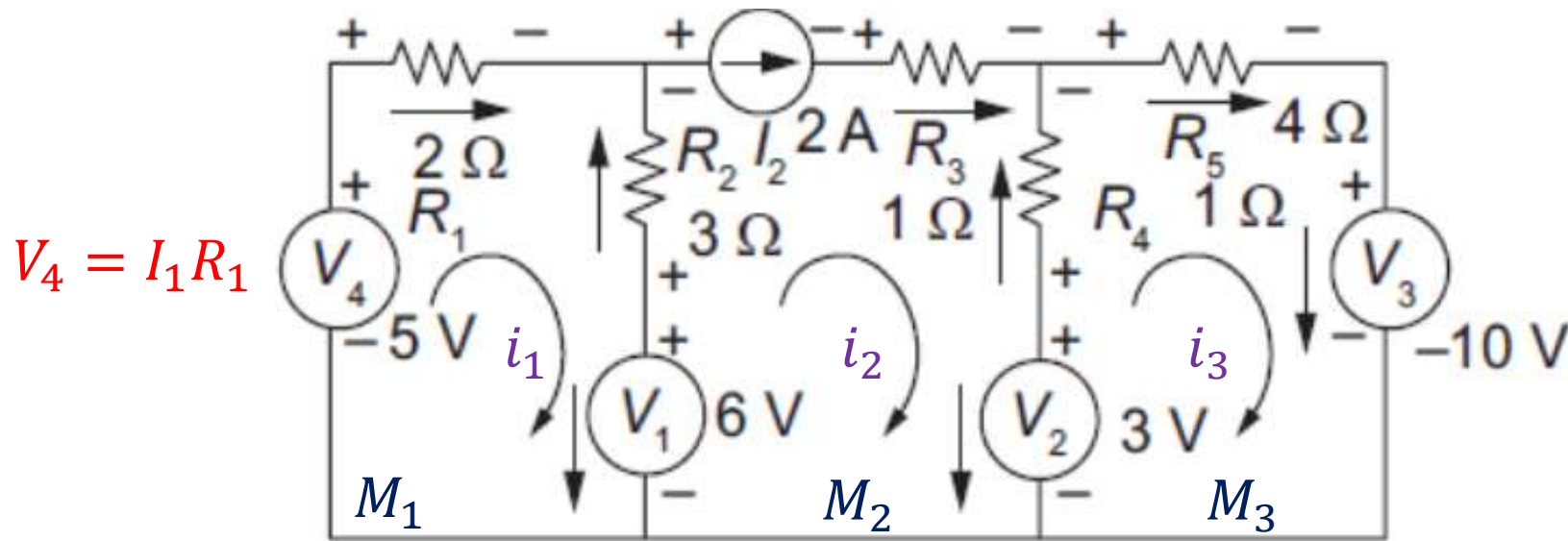
ICS in Series/Parallel with Resistor



ICS in Series/Parallel with Resistor



ICS in Series/Parallel with Resistor



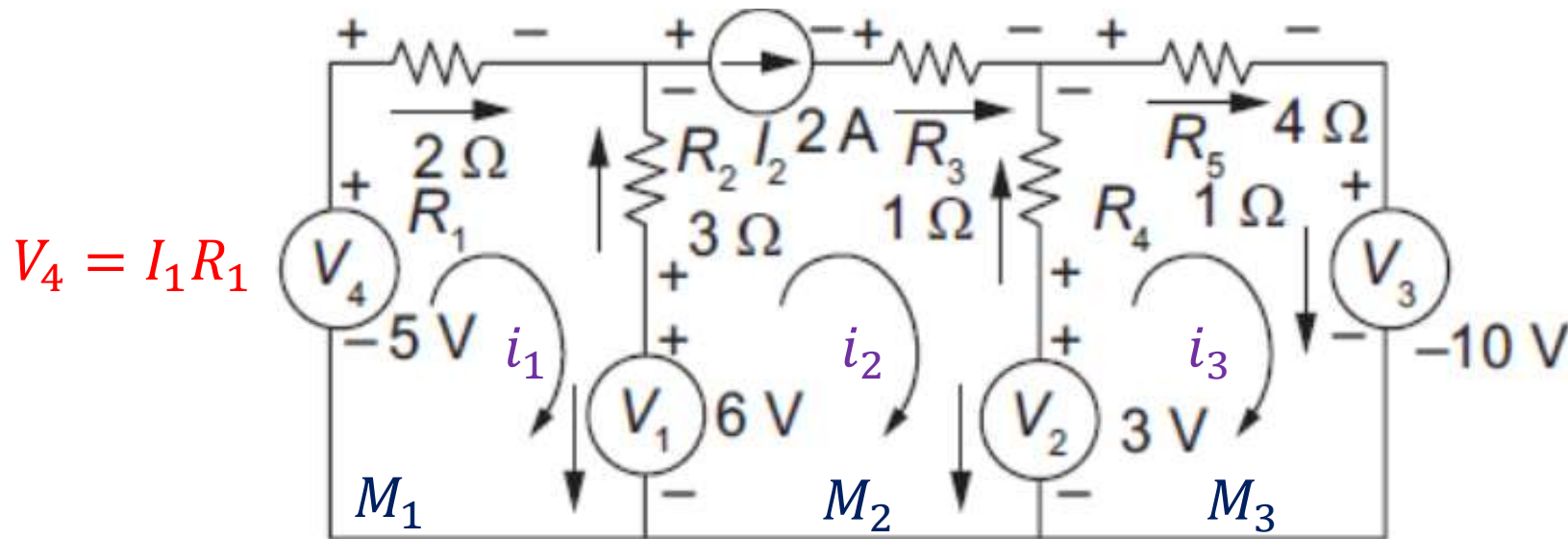
KVL for Mesh **1**

$$-V_4 + R_1 i_1 - R_2 (i_2 - i_1) + V_1 = 0$$

$$(R_1 + R_2) i_1 = R_1 I_1 + R_2 I_2 - V_1$$

$$i_2 = I_2$$

ICS in Series/Parallel with Resistor



KVL for Mesh 3

$$-V_2 + R_4(i_3 - i_2) + R_5 i_3 + V_3 = 0$$

$$(R_4 + R_5)i_3 = R_4 I_2 + V_2 - V_3$$

$$i_2 = I_2$$

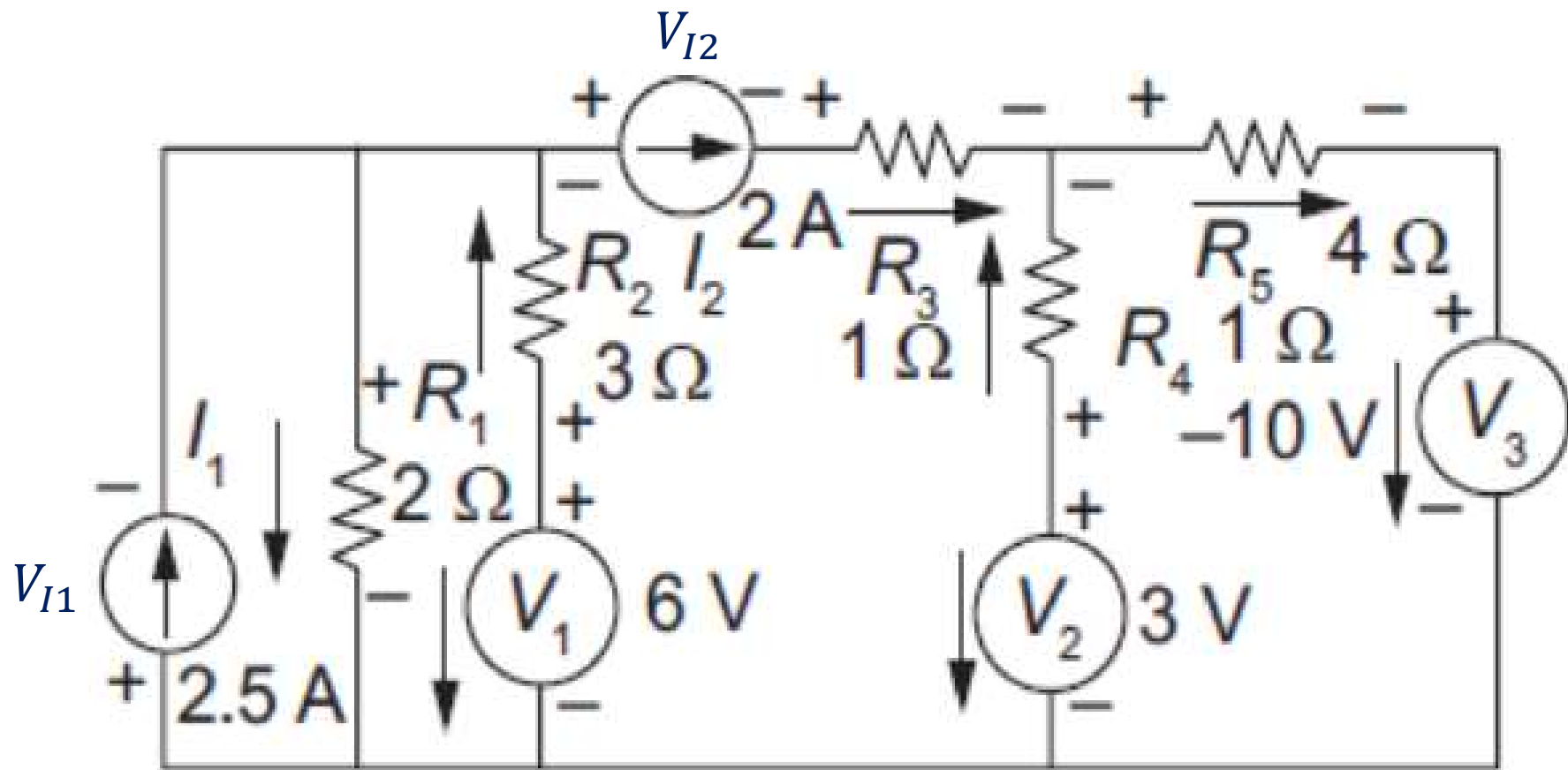
ICS in Series/Parallel with Resistor

$$(R_1 + R_2)i_1 = R_1 I_1 + R_2 I_2 - V_1$$

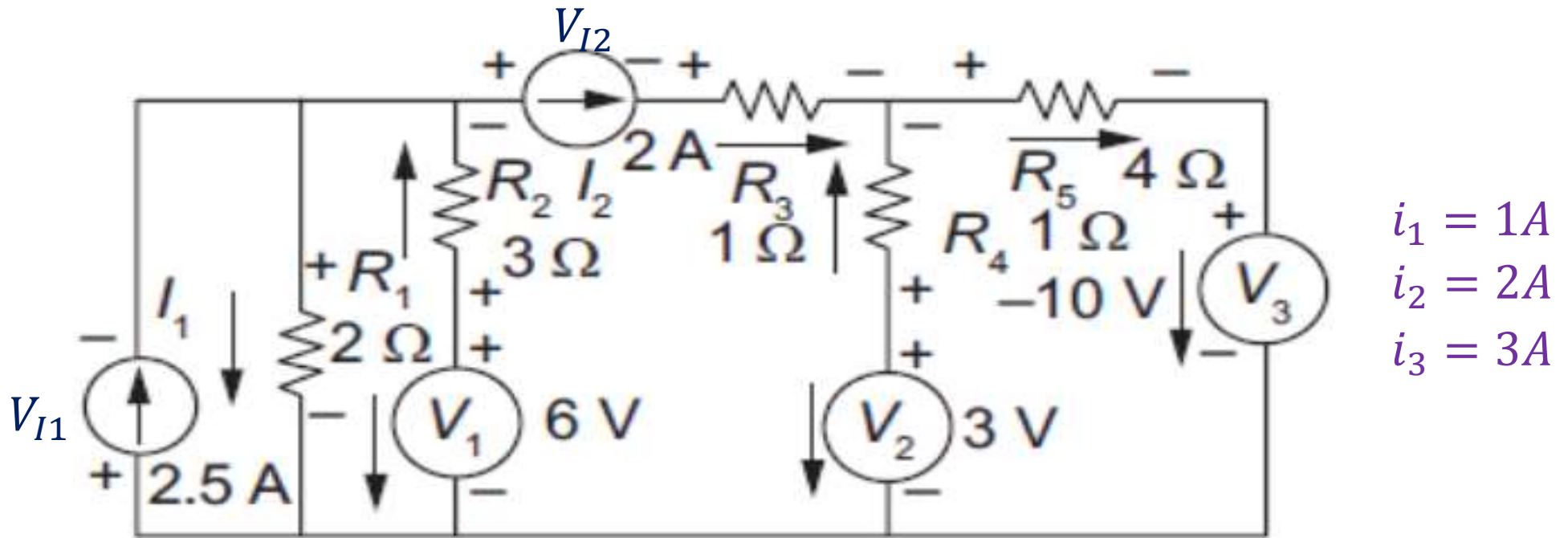
$$(R_4 + R_5)i_3 = R_4 I_2 + V_2 - V_3$$

$$\begin{bmatrix} R_1 + R_2 & 0 \\ 0 & R_4 + R_5 \end{bmatrix} \begin{bmatrix} i_1 \\ i_3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 & R_1 & R_2 \\ 0 & 1 & -1 & 0 & R_4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ I_1 \\ I_2 \end{bmatrix}$$

ICS in Series/Parallel with Resistor

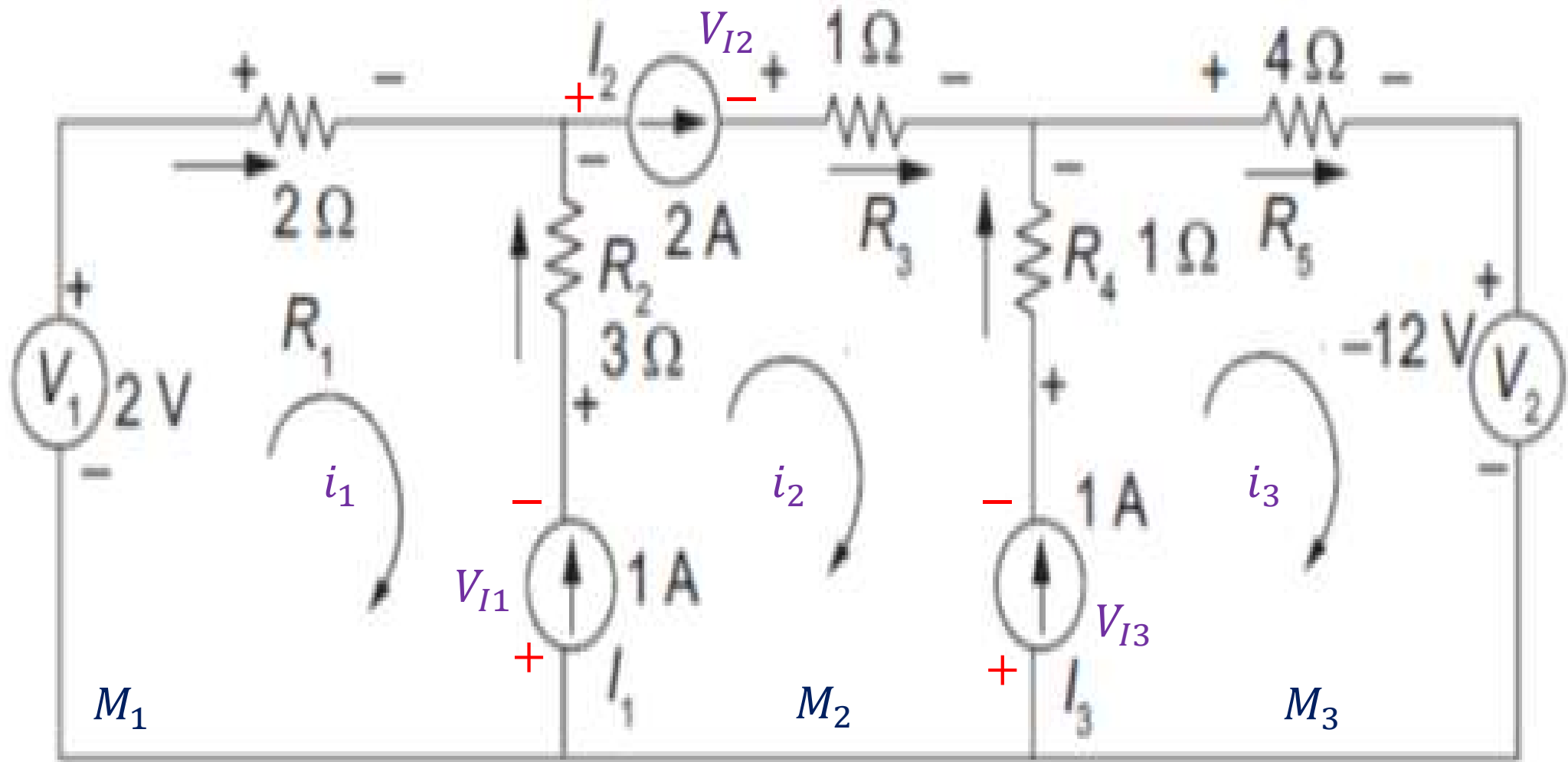


ICS in Series/Parallel with Resistor

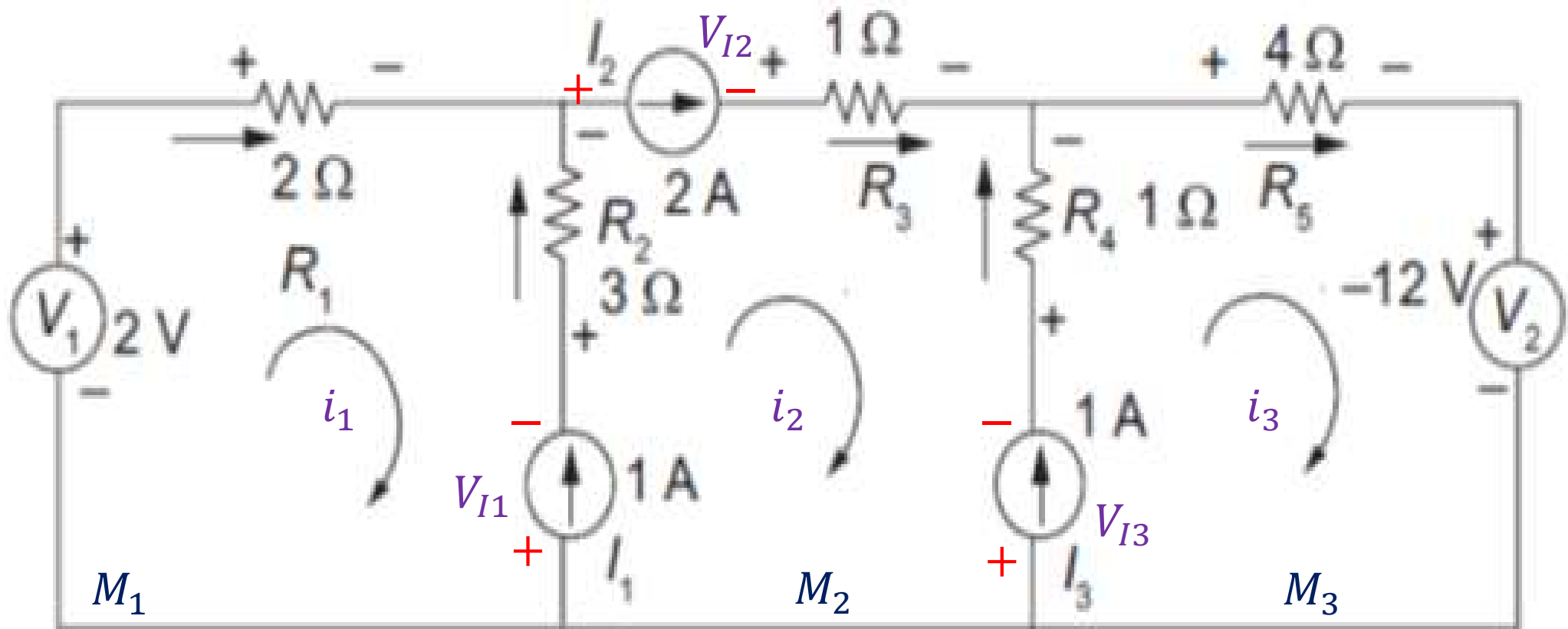


$$-V_1 + R_2(i_2 - i_1) + V_{I2} + R_3 i_2 - R_4(i_3 - i_2) + V_2 = 0$$

ICS Shared between Meshes



ICS Shared between Meshes



$$i_2 = I_2$$

$$i_2 = 2\text{ A}$$

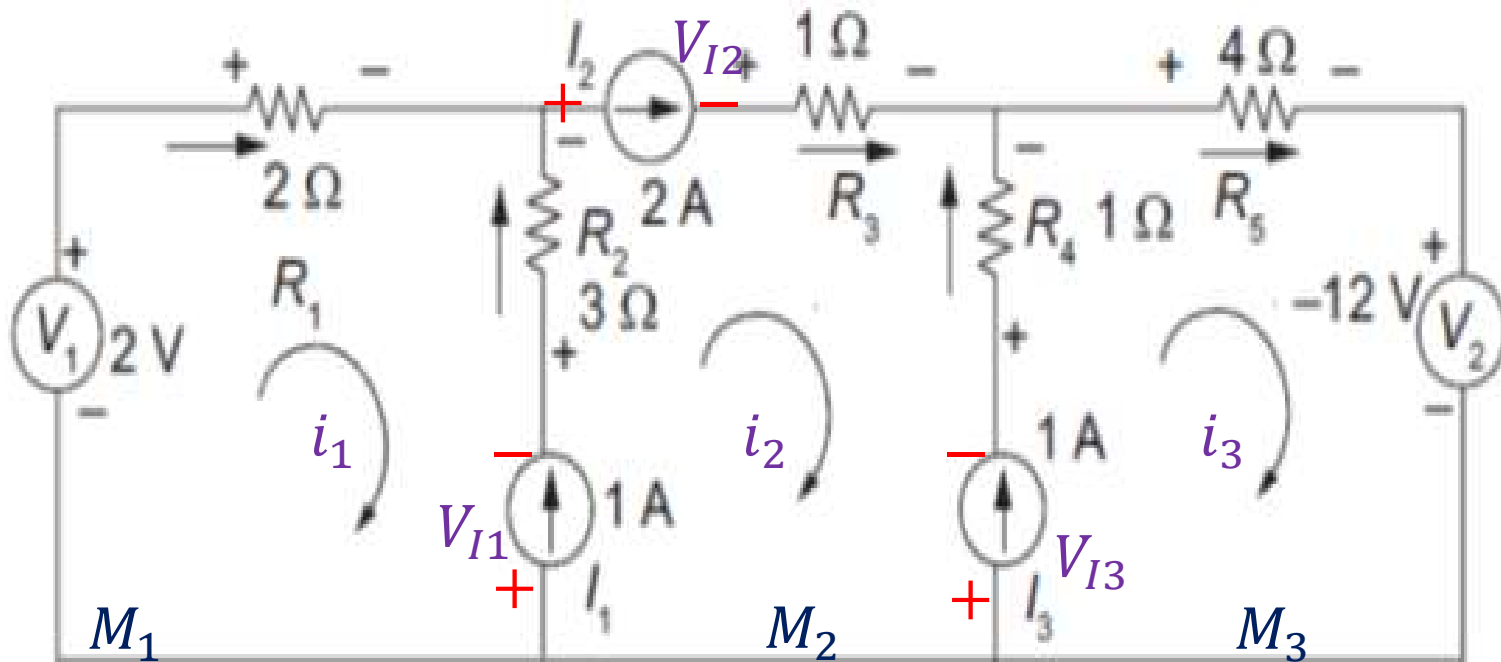
$$i_2 - i_1 = I_1$$

$$i_1 = 1\text{ A}$$

$$i_3 - i_2 = I_3$$

$$i_3 = 3\text{ A}$$

ICS Shared between Meshes

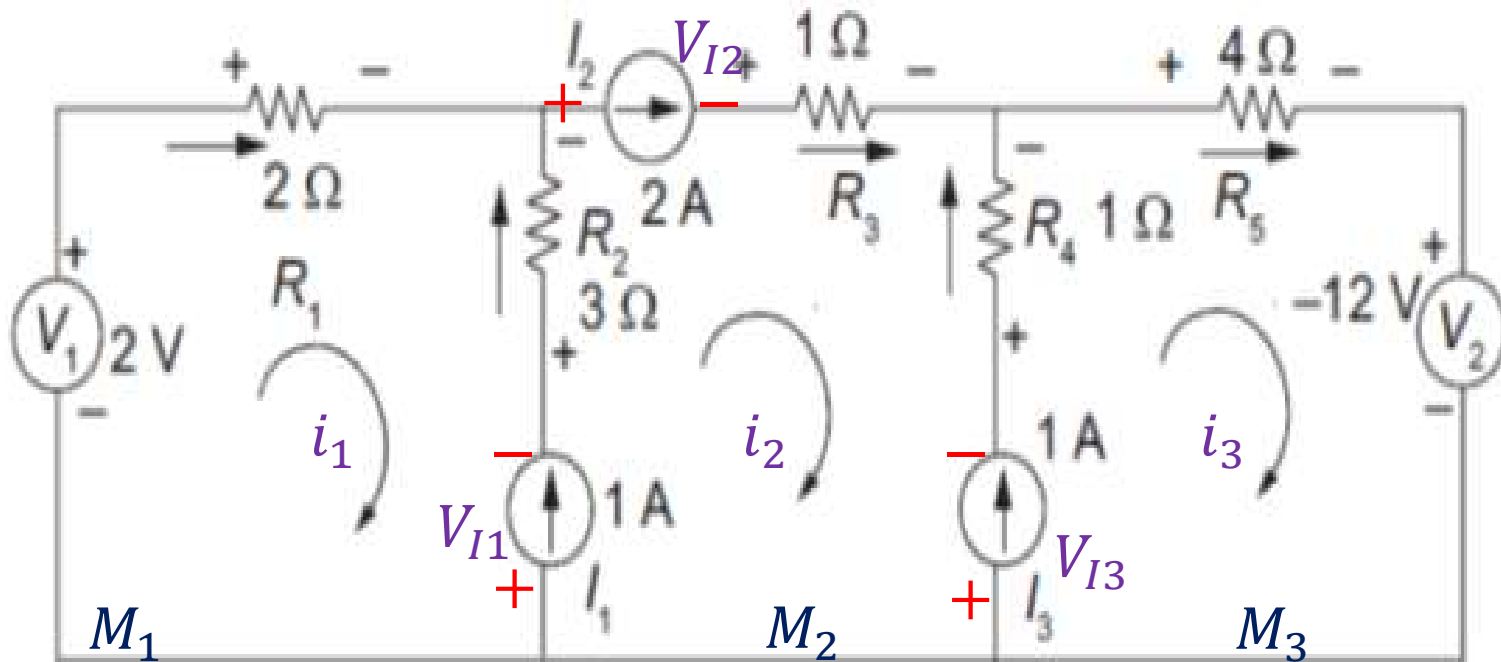


KVL for Mesh **1**

$$-V_1 + R_1 i_1 - R_2 (i_2 - i_1) - V_{I1} = 0$$

$$V_{I1} = -3$$

ICS Shared between Meshes

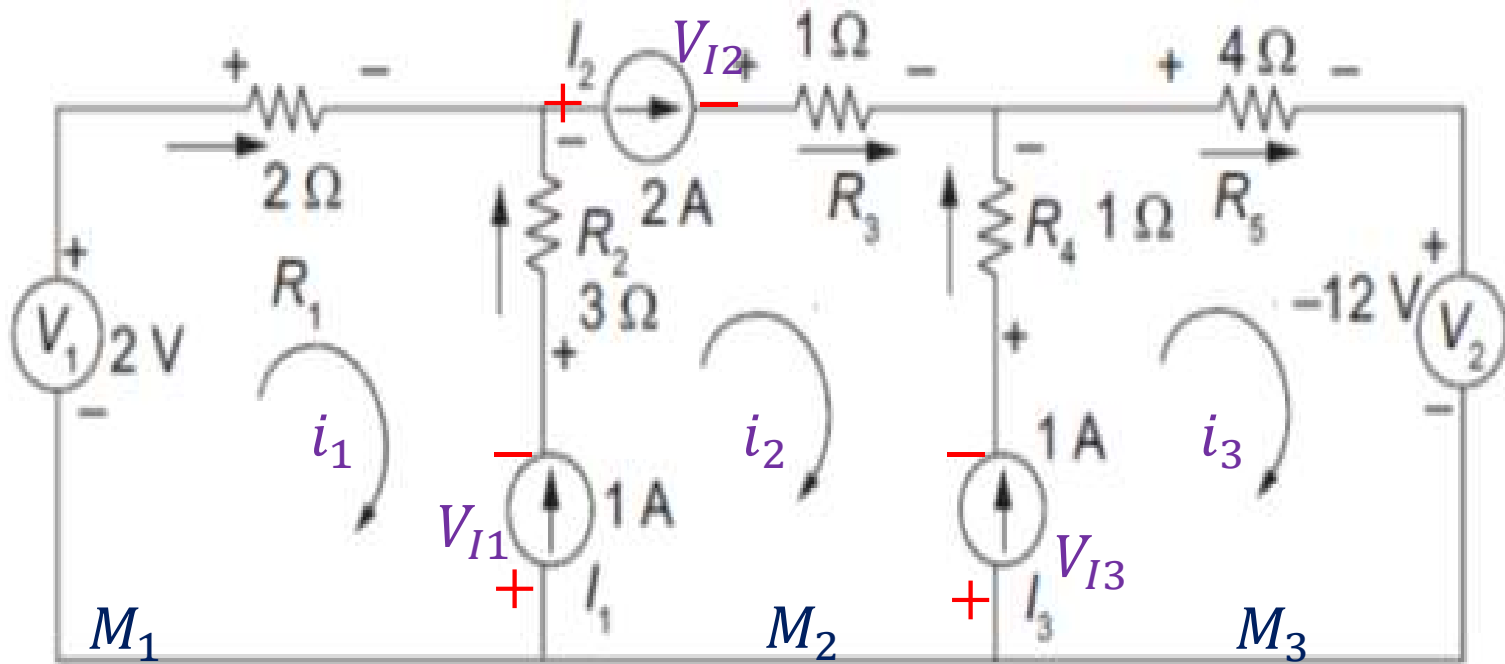


KVL for Mesh 3

$$V_{I3} + R_4(i_3 - i_2) + R_5 i_3 + V_2 = 0$$

$$V_{I3} = -1$$

ICS Shared between Meshes

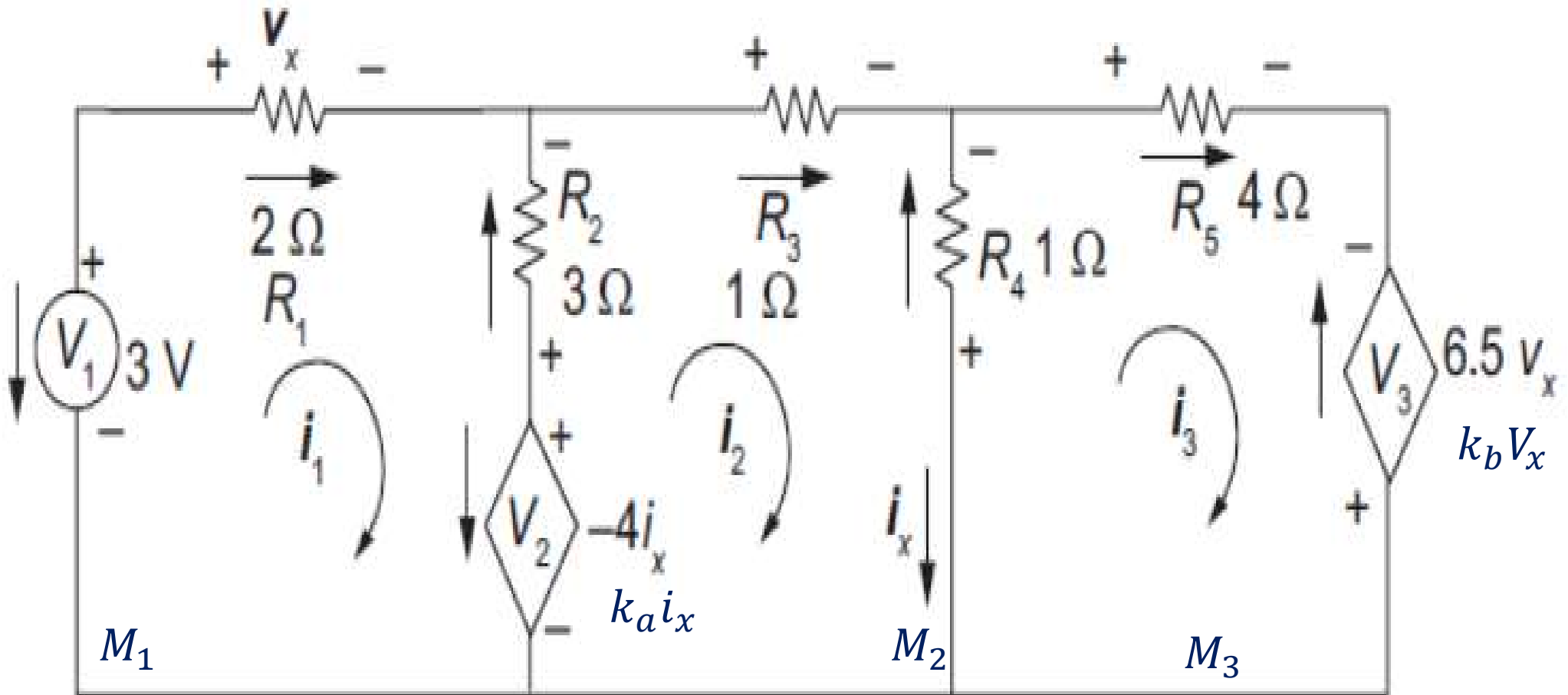


KVL for Mesh 2

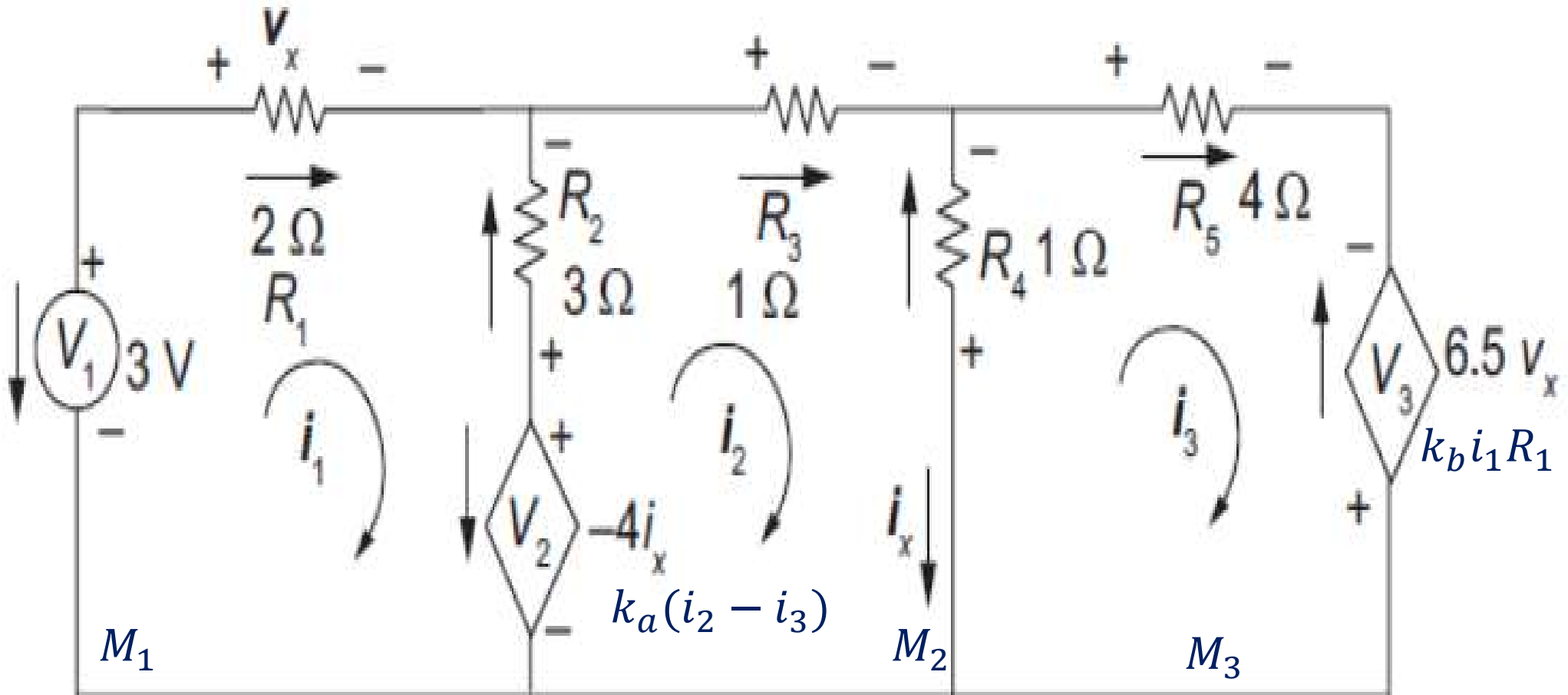
$$V_{I1} + R_2(i_2 - i_1) + V_{I2} + R_3 i_2 - R_4(i_3 - i_2) - V_{I3} = 0$$

$$V_{I2} = -2$$

Mesh Analysis with Dependent Voltage Source



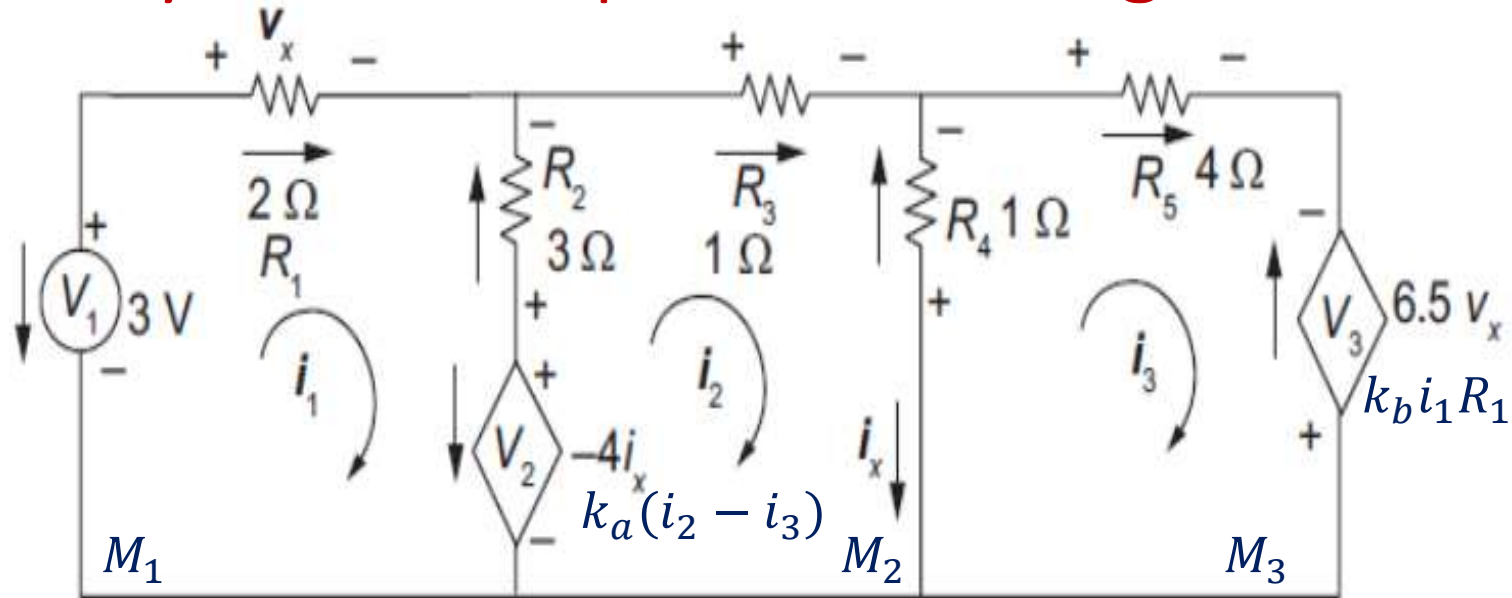
Mesh Analysis with Dependent Voltage Source



$$V_x = i_1 R_1$$

$$i_x = i_2 - i_3$$

Mesh Analysis with Dependent Voltage Source



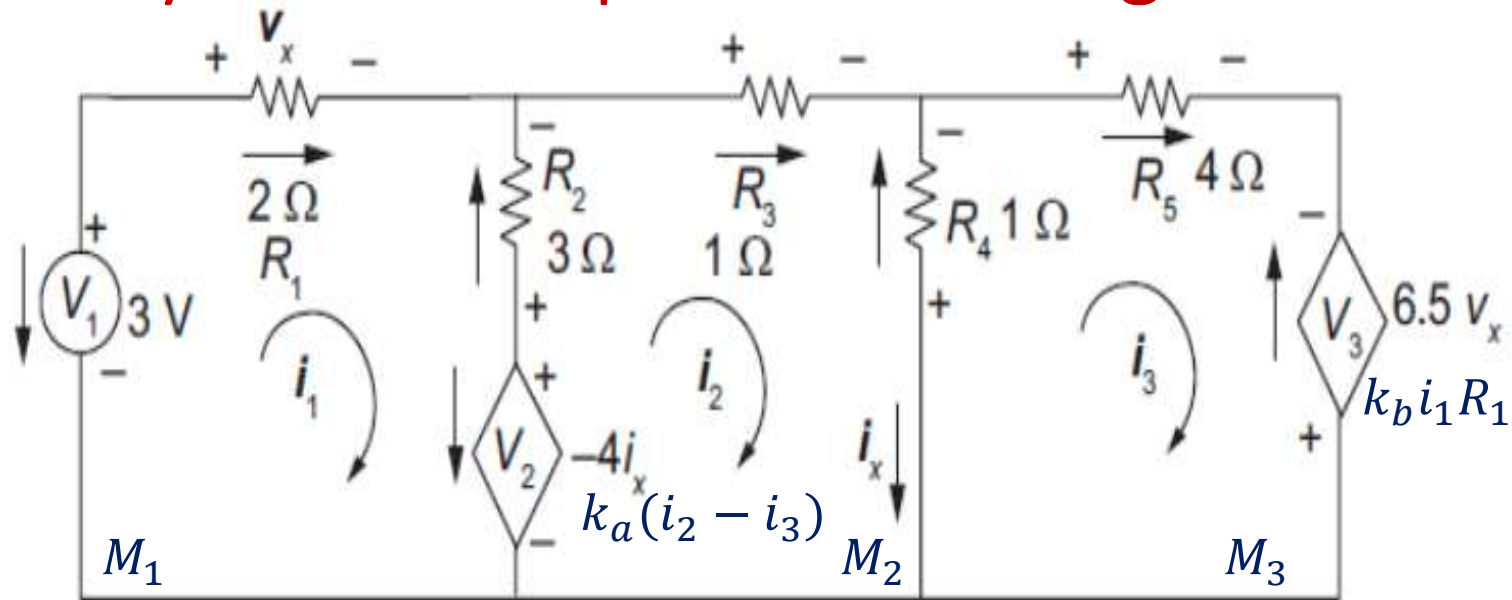
KVL for Mesh **1**

$$-V_1 + R_1 i_1 - R_2 (i_2 - i_1) + V_2 = 0$$

$$-V_1 + R_1 i_1 - R_2 (i_2 - i_1) + k_a (i_2 - i_3) = 0$$

$$(R_1 + R_2) i_1 - (R_2 - k_a) i_2 - k_a i_3 = V_1$$

Mesh Analysis with Dependent Voltage Source



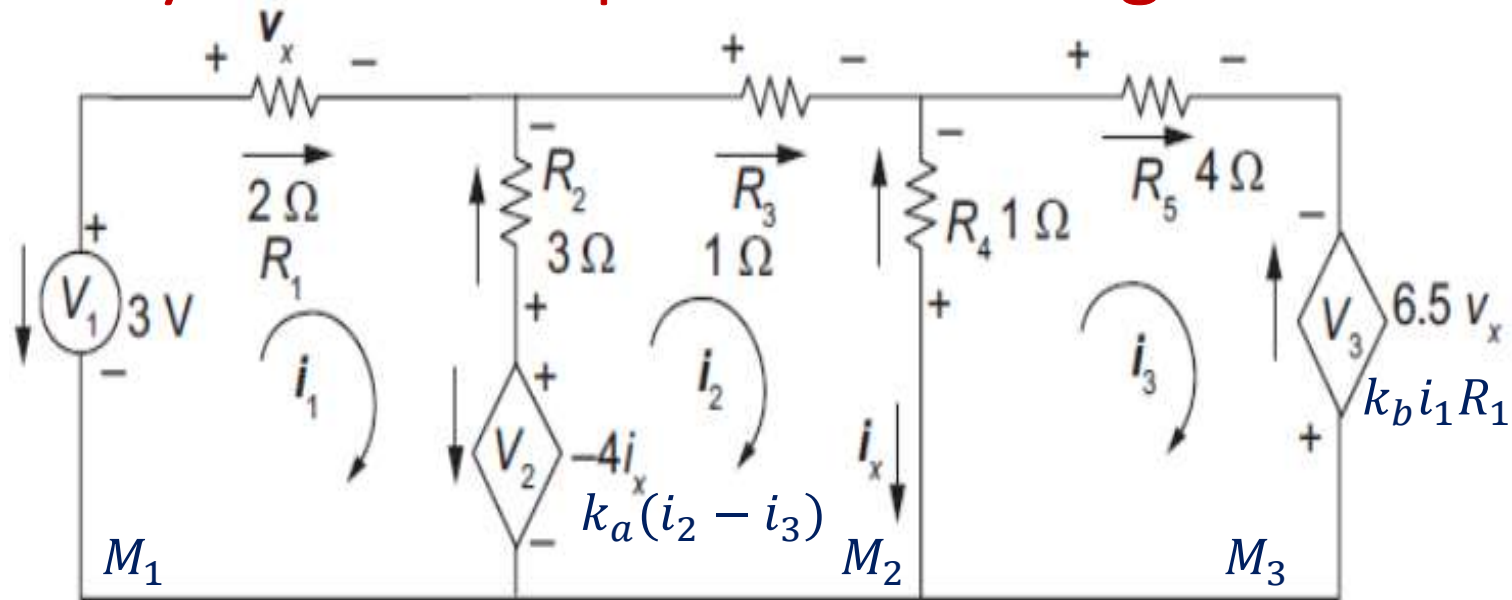
KVL for Mesh **2**

$$-V_2 + R_2(i_2 - i_1) + R_3 i_2 - R_4(i_3 - i_2) = 0$$

$$-k_a(i_2 - i_3) + R_2(i_2 - i_1) + R_3 i_2 - R_4(i_3 - i_2) = 0$$

$$-R_2 i_1 + (R_2 + R_3 + R_4 - k_a) i_2 - (R_4 - k_a) i_3 = 0$$

Mesh Analysis with Dependent Voltage Source



KVL for Mesh **3**

$$R_4(i_3 - i_2) + R_5 i_3 - V_3 = 0$$

$$R_4(i_3 - i_2) + R_5 i_3 - k_b i_1 R_1 = 0$$

$$-k_b R_1 i_1 - R_4 i_2 + (R_4 + R_5) i_3 = 0$$

Mesh Analysis with Dependent Voltage Source

$$(R_1 + R_2)i_1 - (R_2 - k_a)i_2 - k_a i_3 = V_1$$

$$-R_2 i_1 + (R_2 + R_3 + R_4 - k_a)i_2 - (R_4 - k_a)i_3 = 0$$

$$-k_b R_1 i_1 - R_4 i_2 + (R_4 + R_5)i_3 = 0$$

$$\begin{bmatrix} R_1 + R_2 & -(R_2 - k_a) & -k_a \\ -R_2 & R_2 + R_3 + R_4 - k_a & -(R_4 - k_a) \\ -k_b R_1 & -R_4 & R_4 + R_5 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} [V_1]$$

Summary

- Mesh Analysis with Independent Voltage Sources
- Mesh Analysis with Independent Current Sources
 - Current Source in Parallel with Resistor
 - Current Source in Series with Resistor
 - Current Source Shared between Meshes
- Mesh Analysis with Dependent Sources



Thank You